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CONTROLLED CHIRP IN OPTICAL DEVICES

Abstract

An electro-optic device including a first waveguide, a second waveguide, and electrodes is described. The first waveguide includes a first thin film lithium-containing (TFLC) electro-optic material and carries a first optical signal. The second waveguide includes a second TFLC electro-optic material and carries a second optical signal. The electrodes include a differential electrode pair proximate to a portion of the first waveguide and to a portion of the second waveguide. The differential electrode pair is configured to provide a first modulation to the first optical signal and a second modulation to the second optical signal. A first magnitude of the first modulation is different from a second magnitude of the second modulation such that the electro-optic device has an engineered chirp.

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Background/Summary

CROSS REFERENCE TO OTHER APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 18/972,332 entitled CONTROLLED CHIRP IN OPTICAL DEVICES filed Dec. 6, 2024, which claims priority to U.S. Provisional Patent Application No. 63/607,485 entitled CONTROLLED CHIRP IN OPTICAL DEVICES filed Dec. 7, 2023, both which are incorporated herein by reference for all purposes.

BACKGROUND OF THE INVENTION

[0002] Optical modulators may be used to encode data into optical signals (e.g. visible, infrared, or other portions of the spectrum of light). For example, in an electro-optic modulator, an electrical signal (e.g. having a frequency in the microwave range—from hundred of kHz through hundreds of GHz) is driven through electrodes that are in proximity to a waveguide. The electric field generated by the electrical signal in the electrodes (also termed “electrode signal” herein) changes the index of refraction of an electro-optic material that carries the optical signal. The change in index of refraction can provide intensity or phase modulation, depending on the configuration of the waveguide. Thus, data may be optically transmitted.

[0003] Although signals may be transmitted optically, the optical signal may be subject to chirp. Chirp may occur when a modulator includes two waveguide arms, each of which undergoes modulation as described herein. If there are differences in the modulation (e.g. the magnitude of intensity modulation), chirp may be introduced. Chirp may be undesirable in optical signals because optical signals are generally transmitted via optical fibers. Optical fibers are often subject to dispersion for certain wavelengths. For example, some optical fibers are dispersion free at a particular wavelength (e.g. 1311 nanometers), but introduce positive or negative dispersion for higher or lower wavelengths. As a result, the optical signal may be distorted. Accordingly, what is desired are techniques for reducing distortion in optical signals.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] Various embodiments of the invention are disclosed in the following detailed description and the accompanying drawings.

[0005] FIGS. 1A-1C depict embodiments of electro-optic devices having engineered chirp.

[0006] FIG. 2 depicts an embodiment of an electro-optic device having engineered chirp.

[0007] FIG. 3 depicts an embodiment of an electro-optic device having engineered chirp.

[0008] FIG. 4 depicts an embodiment of an electro-optic device having engineered chirp.

[0009] FIG. 5 depicts an embodiment of an electro-optic device having engineered chirp.

[0010] FIG. 6 depicts an embodiment of an electro-optic device having engineered chirp.

[0011] FIG. 7 is a flow chart depicting an embodiment of a method for using an electro-optic device having engineered chirp.

DETAILED DESCRIPTION

[0012] The invention can be implemented in numerous ways, including as a process; an apparatus; a system; a composition of matter; a computer program product embodied on a computer readable storage medium; and/or a processor, such as a processor configured to execute instructions stored on and/or provided by a memory coupled to the processor. In this specification, these implementations, or any other form that the invention may take, may be referred to as techniques.

In general, the order of the steps of disclosed processes may be altered within the scope of the invention. Unless stated otherwise, a component such as a processor or a memory described as being configured to perform a task may be implemented as a general component that is temporarily configured to perform the task at a given time or a specific component that is manufactured to perform the task. As used herein, the term 'processor' refers to one or more devices, circuits, and/or processing cores configured to process data, such as computer program instructions.

[0013] A detailed description of one or more embodiments of the invention is provided below along with accompanying figures that illustrate the principles of the invention. The invention is described in connection with such embodiments, but the invention is not limited to any embodiment. The scope of the invention is limited only by the claims and the invention encompasses numerous alternatives, modifications and equivalents. Numerous specific details are set forth in the following description in order to provide a thorough understanding of the invention. These details are provided for the purpose of example and the invention may be practiced according to the claims without some or all of these specific details. For the purpose of clarity, technical material that is known in the technical fields related to the invention has not been described in detail so that the invention is not unnecessarily obscured.

[0014] Although data may be transmitted via optical signals, the optical signal may be subject to noise or other issues that affect transmission. For example, chirp may be inadvertently introduced during modulation. Signals traveling through optical fibers also experience dispersion. This dispersion in combination with the uncontrolled chirp may result in a distortion of the optical signal that interferes with or prevents the receiver from properly decoding the information carried by the optical signal. Accordingly, what is desired are techniques for reducing distortion in optical signals.

[0015] An electro-optic device includes a first waveguide, a second waveguide, and electrodes. The first waveguide includes a first thin film lithium-containing (TFLC) electro-optic material and is configured to carry a first optical signal. The second waveguide includes a second TFLC electro-optic material and is configured to carry a second optical signal. The plurality of electrodes includes a differential electrode pair proximate to a portion of the first waveguide and to a portion of the second waveguide. The differential electrode pair is configured to provide a first modulation to the first optical signal and a second modulation to the second optical signal. A first magnitude of the first modulation is different from a second magnitude of the second modulation such that the electro-optic device has an engineered chirp.

[0016] In some embodiments, the electrodes include a first signal electrode, a second signal electrode, and a ground electrode. The first signal electrode and the second signal electrode form the differential electrode pair. The first signal electrode, the second signal electrode, and the ground electrode form a differential electrode trio. In some embodiments, the first signal electrode and the second signal electrode are separated by a first gap proximate to the first waveguide. In such embodiments, the second signal electrode and the ground electrode are separated by a second gap proximate to the second waveguide. The second gap is different from the first gap such that the first modulation has the first magnitude and the second modulation has the second magnitude. In some embodiments, the first gap and the second gap are configured such that the first magnitude ($M1$) for the first waveguide is twice as large as the second magnitude ($M2$) for the second waveguide, electro-optic device having chirp parameter (C) of $C=1/3$, with $C=(M1-M2)/(M1+M2)$. In some embodiments, the first gap and the second gap are configured such that the first magnitude ($M1$) for the first waveguide and the second magnitude ($M2$) for the second waveguide result in the electro-optic device having a chirp parameter of at least 0.1 and not greater than 0.9 or not greater than -0.1 and at least -0.9, where $C=(M1-M2)/(M1+M2)$ ($-0.9 \leq C \leq -0.1$ or $0.1 \leq C \leq 0.9$).

[0017] In some embodiments, the electro-optic device includes third and fourth waveguides including third and fourth TFLC electro-optic materials, respectively, and configured to carry third and fourth optical signals, respectively. In such embodiments, the electrodes include a second differential electrode pair proximate to a portion of the third waveguide and to a portion of the

fourth waveguide. The differential electrode pair has a first length proximate to the first and second waveguides and the second differential electrode pair has a second length proximate to the third and fourth waveguides. The first length is different from the second length.

[0018] In some embodiments, the first waveguide and the second waveguide are coupled to an optical signal transmitter having a dispersion. For example, the optical transmitter may include an optical fiber coupled to the first and second waveguides and having the dispersion. The engineered chirp is opposite to the dispersion. In some embodiments, the first and second waveguides are arms of a modulator waveguide including the first waveguide, the second waveguide, a first splitter and a second splitter. The first and second waveguides are between the first splitter and the second splitter.

[0019] An electro-optic device including waveguide pairs and electrode trios is described. Each of the waveguide pairs includes a first waveguide and a second waveguide. Each of the waveguide pairs including a thin film lithium-containing (TFLC) electro-optic material and is configured to carry a first optical signal in the first waveguide and a second optical signal in the second waveguide. Each of the electrode trios includes a differential electrode pair and a ground electrode. A first signal electrode of the differential electrode pair is proximate to a portion of the first waveguide. A second signal electrode of differential electrode pair is proximate to a region between the first waveguide and the second waveguide. The ground electrode is proximate to a portion of the second waveguide. Each of the electrode trios is configured to provide a first modulation having a first magnitude to the first optical signal and a second modulation having a second magnitude to the second optical signal. The first magnitude is different from the second magnitude for at least one of the plurality of electrode trios such that at least one of the plurality of waveguide pairs has an engineered chirp.

[0020] In some embodiments, the first signal electrode and the second signal electrode of the electrode trio(s) are separated by a first gap proximate to the first waveguide of the waveguide pair(s). The second signal electrode and the ground electrode of the electrode trio(s) are separated by a second gap proximate to the second waveguide of the waveguide pair(s). The second gap is different from the first gap such that the first modulation has the first magnitude and the second modulation has the second magnitude for the waveguide pair(s). In some embodiments, the first gap and the second gap for the electrode trio(s) are configured such that the first magnitude (M1) and the second magnitude (M2) for the first and second waveguides of the waveguide pair(s) result in of the waveguide pair(s) has a chirp parameter of at least 0.1 and not greater than 0.9 or not greater than -0.1 and at least -0.9, where $C=(M1-M2)/(M1+M2)$.

[0021] In some embodiments, a first electrode trio has a first length proximate to a first waveguide pair and a second electrode trio has a second length proximate to a second waveguide pair. The first length being different from the second length. Each of the waveguide pairs may be coupled to an optical signal transmitter having a dispersion. The engineered chirp is configured based on the dispersion. In some embodiments, the optical signal transmitter includes an optical fiber coupled to the electro-optic device.

[0022] A method includes providing a first optical signal to a first waveguide including a first thin film lithium-containing (TFLC) electro-optic material. A second optical signal is provided to a second waveguide including a second TFLC electro-optic material. An electro-optic device includes the first waveguide and the second waveguide. Electrode signals are driven through electrodes. The electrodes include a differential electrode pair proximate to a portion of the first waveguide and proximate to a portion of the second waveguide. The electrode signals in the differential electrode pair are configured to provide a first modulation having a first magnitude to the first optical signal and a second modulation having a second magnitude to the second optical signal. The first modulation being different from the second modulation such that the electro-optic device has an engineered chirp.

[0023] In some embodiments, the electrodes include a first signal electrode, a second signal

electrode, and a ground electrode. The differential electrode pair includes the first signal electrode and the second signal electrode. The first signal electrode, the second signal electrode, and the ground electrode form a differential electrode trio. In some embodiments, the first signal electrode and the second signal electrode are separated by a first gap proximate to the first waveguide. The second signal electrode and the ground electrode may be separated by a second gap proximate to the second waveguide. The second gap is different from the first gap such that the first magnitude (M1) for the first waveguide and the second magnitude (M2) for the second waveguide result in the electro-optic device having a chirp parameter of at least 0.1 and not greater than 0.9 or not greater than -0.1 and at least -0.9, where $C=(M1-M2)/(M1+M2)$.

[0024] In some embodiments, the method includes driving third and fourth optical signals through third and fourth waveguides, respectively. The third waveguide includes a third TFLC electro-optic material. The fourth waveguide includes a fourth TFLC electro-optic material. In such embodiments, the electrodes includes a second differential electrode pair proximate to a portion of the third waveguide and to a portion of the fourth waveguide. The differential electrode pair has a first length proximate to the first and second waveguides. The second differential electrode pair has a second length proximate to the third and fourth waveguides. The first length is different from the second length. In some embodiments, the first and second waveguides are coupled to an optical signal transmitter having a dispersion. The engineered chirp is opposite to the dispersion.

[0025] The techniques may be described in the context of positive and negative electrodes, positive and negative electrical signals (e.g., voltages), positive and negative inputs, and positive and negative outputs. However, such electrodes, voltages, inputs, and outputs carry or are signals that are opposite in polarity with respect to a reference. Stated differently, positive and negative refer to polarity with respect to a reference. In some embodiments, the reference is ground. In some embodiments, the reference is a nonzero voltage. In some embodiments, the reference is constant. In some embodiments, the reference may be slowly varying. In such embodiments, the positive voltage, or signal, has the opposite polarity with respect to the reference as the negative voltage, or signal. Thus, the terms “positive” and “negative” simply indicate that the signals are opposite in polarity with respect to the reference. There is no requirement that the “positive” signal remain positive with respect to the reference or that the “negative” signal remains negative with respect to the reference. Further, although described as ground, in some embodiments, the ground may include a slowly varying voltage. In addition, various features of the electro-optic devices are described herein. One or more of these features may be combined in manners not explicitly described herein.

[0026] FIGS. 1A-1C depict embodiments of electro-optic devices **100** and **100'** having engineered chirp. FIG. 1A depicts a plan view of a portion of electro-optic device **100**, while FIG. 1B depicts a cross-sectional view of a portion of electro-optic device **100**. FIG. 1C depicts a plan view of portion of electro-optic device **100'**. For clarity, not all components are shown and FIGS. 1A-1C are not to scale. FIGS. 1A-1C may be considered to show portions of electro-optic devices **100** and **100'**. For example, drivers, voltage sources, the source of the optical signal, and other portions of such a system are not shown.

[0027] Referring to FIGS. 1A-1B, electro-optic device **100** includes electrodes **120**, waveguide **130**, and cladding **140** that reside on substrate structure **101**. Waveguide **130** includes arms **132** and **134**. In some embodiments, arms **132** and **134** may be considered separate waveguides. Waveguide **130** may be configured as a Mach-Zehnder interferometer. Electrodes **120** include ground electrode **126** and an electrode trio **124**. Electrode trio **124** includes differential electrodes **124+** and **124-** and ground electrode **124G**. Thus, differential electrodes **124+** and **124-** may carry signals that are opposite in polarity (e.g. one hundred and eighty degrees out of phase). Differential electrode **124+** may be considered to carry the positive voltage signal, while differential electrode **124-** carries a negative voltage signal (i.e., a signal of opposite polarity to the positive voltage signal). The frequency of the electrical signals carried by differential electrodes **124+** and **124-** may be high, for

example in the 100 kHz through 100 GHz (or higher) range. Ground electrode **124G** may also have a slow varying potential or a constant bias (zero or nonzero) voltage. Waveguide arm **132** is between differential electrodes **124+** and **124-**, while waveguide arm **134** is between signal electrode **124-** and ground electrode **124G**. Electrode **124-** is depicted as being between waveguide arms **132** and **134**. In some embodiments, two electrodes carrying the same signal may be used instead of a single electrode **124-**.

[0028] FIG. **1B** depicts a cross-sectional view of electro-optic device **100**. Each waveguide arm **132** and **134** includes ridge portion **133** and slab portion **135**. In some embodiments, slab portion **135** may be omitted from arm **132** and/or **134**. The optical modes carried by waveguide arms **132** and **134** are generally well confined to ridge portion **133**. Also shown are electrode trio **124** (positive electrode **124+**, negative electrode **124-** and ground electrode **124G**). In some embodiments, extensions **122** may extend from differential electrodes **124**. Although not shown in FIG. **1B**, in some embodiments, ground electrode **124G** may include extensions. Extensions **122** may allow for accumulation of charges closer to arms **132** and **134** (e.g., ridge **133**) of waveguide **130**. These charges are due to the electrode signals carried by differential electrodes **124+** and **124-**. Thus, extensions **122** may enhance the electric field experienced by ridge **133** and improve modulation of the optical signal carried by arms **132** and **134**. Embodiments of analogous electrodes may be found in co-pending U.S. patent application Ser. No. 17/843,906, entitled ELECTRO-OPTIC DEVICES HAVING ENGINEERED ELECTRODES, which is a continuation of U.S. patent application Ser. No. 17/102,047 entitled ELECTRO-OPTIC DEVICES HAVING ENGINEERED ELECTRODES, filed Nov. 23, 2020, which claims priority to U.S. Provisional Patent Application No. 62/941,139 entitled THIN-FILM ELECTRO-OPTIC MODULATORS filed Nov. 27, 2019, U.S. Provisional Patent Application No. 63/033,666 entitled HIGH PERFORMANCE OPTICAL MODULATORS filed Jun. 2, 2020, and U.S. Provisional Patent Application No. 63/112,867 entitled BREAKING VOLTAGE-BANDWIDTH LIMIT IN INTEGRATED LITHIUM NIOBATE MODULATORS USING MICRO-STRUCTURED ELECTRODES filed Nov. 12, 2020, all of which are incorporated herein by reference for all purposes.

[0029] Also shown in FIG. **1B** is substrate structure **101**. In some embodiments, substrate structure **101** includes an underlying substrate (e.g., silicon) and a thick buried oxide underlayer between slab **135** and the underlying substrate. For example, the buried oxide underlayer may be at least three micrometers thick, at least five micrometers thick, at least seven micrometers thick, and not more than fifteen micrometers thick. In some embodiments, the buried oxide layer includes or consists of silicon dioxide. Other components may be present in substrate structure **101** but are not shown for simplicity.

[0030] Referring to FIGS. **1A-1B**, waveguide **130** (and thus each arm **132** and **134**) includes at least one optical material possessing an electro-optic effect. In some embodiments, waveguide **130** includes thin film lithium containing (TFLC) electro-optic materials, such as thin film lithium niobate (TFLN) and/or thin film lithium tantalate (TFLT). In some embodiments, waveguide **130** consists of TFLN and/or TFLT. For example, the thickness of the electro-optic layer(s) from which waveguide **130** are formed is not more than three micrometers prior to fabrication of waveguides **130**. In some embodiments, this thickness is not more than 1.5 micrometer or not more than one micrometer. In some embodiments, the thickness is not more than seven hundred nanometers or not more than five hundred nanometers. In some embodiments, the thickness is at least one hundred nanometers. Other thicknesses are possible. The thickness of waveguides **130** is less than or equal to the thickness of the as-provided electro-optic layer. For example, the thickness of waveguides **130** may be on the order of a few hundred nanometers or less.

[0031] Although primarily described in the context of (TFLC) electro-optic materials, such as thin film lithium niobate (TFLN) and thin film lithium tantalate (TFLT), other nonlinear optical materials may be used in the optical devices described herein. For example, other ferroelectric

nonlinear (e.g. second order) optical materials may also be desired to be used in waveguides **130**. Such ferroelectric nonlinear optical materials may include but are not limited to potassium niobate (e.g. KNbO_3), gallium arsenide (GaAs), potassium titanyl phosphate (KTP), lead zirconate titanate (PZT), and barium titanate (BaTiO_3). The techniques described may also be used for other nonlinear ferroelectric optical materials, particularly those which may otherwise be challenging to fabricate. For example, such nonlinear ferroelectric optical materials may have inert chemical etching reactions using conventional etching chemicals such as fluorine, chlorine or bromine compounds.

[0032] In some embodiments, the optical material(s) used in waveguide **130** are nonlinear. As used herein, a nonlinear optical material exhibits the electro-optic effect and has an effect that is at least (e.g. greater than or equal to) 5 picometer/volt. In some embodiments, the nonlinear optical material has an effect that is at least 10 picometer/volt. In some such embodiments nonlinear optical material has an effect of at least 20 picometer/volt. The nonlinear optical material experiences a change in index of refraction in response to an applied electric field. In some embodiments, the nonlinear optical material is ferroelectric. In some embodiments, the electro-optic material effect includes a change in index of refraction in an applied electric field due to the Pockels effect. Thus, in some embodiments, optical materials possessing the electro-optic effect in one or more the ranges described herein are considered nonlinear optical materials regardless of whether the effect is linearly or nonlinearly dependent on the applied electric field. The nonlinear optical material may be a non-centrosymmetric material. Therefore, the nonlinear optical material may be piezoelectric. Such nonlinear optical materials may have inert chemical etching reactions for conventional etching using chemicals such as fluorine, chlorine or bromine compounds. In some embodiments, the nonlinear optical material(s) include one or more of LN, LT, potassium niobate, gallium arsenide, potassium titanyl phosphate, lead zirconate titanate, and barium titanate. In other embodiments, other nonlinear optical materials having analogous optical characteristics may be used.

[0033] In some embodiments, waveguide arms **132** and **134** are low optical loss waveguides. For example, waveguide arms **132** and **134** may each have a total optical loss of not more than 10 dB through the portion of waveguide **130** (e.g. when biased at maximum transmission and as a maximum loss) in proximity to differential electrodes **124+** and **124-**. The total optical loss is the optical loss in a waveguide through a single continuous electrode region (e.g. as opposed to multiple devices cascaded together). In some embodiments, waveguide **130** has a total optical loss of not more than 8 dB. In some embodiments, the total optical loss is not more than 4 dB. In some embodiments, the total optical loss is less than 3 dB. In some embodiments, the total optical loss is less than 2 dB. In some embodiments, waveguides **130** has an optical loss of not more than 3 dB/cm (e.g. on average). In some embodiments, the nonlinear material(s) in waveguides **130** has an optical loss of not more than 2.0 dB/cm. In some such embodiments, waveguide **130** has an optical loss of not more than 1.0 dB/cm. In some embodiments, waveguide **130** has an optical loss of not more than 0.5 dB/cm. In some embodiments, the low optical losses are associated with a low surface roughness of the side walls of waveguides **130**.

[0034] Waveguide **130**, and thus waveguide arms **132** and **134**, may have improved surface roughness. For example, the short range root mean square surface roughness of a sidewall of the ridge **133** may be less than ten nanometers. In some embodiments, this root mean square surface roughness is not more than five nanometers. In some cases, the short range root mean square surface roughness does not exceed two nanometers. In some embodiments, the height of ridge **133** is selected to provide a confinement of the optical mode such that there is a 10 dB reduction in intensity from the intensity at the center of ridge **133** at ten micrometers from the center of ridge **133**. For example, the height of ridge **133** is on the order of a few hundred nanometers in some cases. However, other heights are possible in other embodiments. Various other optical components may be incorporated into waveguide **130** to provide the desired functionality. For example,

waveguide **130** may have wider portion(s) (not shown in FIGS. **1A-1B**) for accommodating multiple modes.

[0035] In operation, an optical signal is provided to waveguide **130**. The optical signal is split and provided to waveguide arms **132** and **134**. In some embodiments, optical signals are evenly split between arms **132** and **134** (e.g. half of the intensity to each arm **132** and **134**). In other embodiments, the optical signals may be split differently. Differential electrical signals (e.g., complementary signals) are provided to differential electrodes **124+** and **124-**. These differential signals may be one hundred and eighty degrees out-of-phase. The differential electrode signals carried by differential electrodes **124+** and **124-** modulate the optical signal in waveguide arm **132** in regions where electrodes **124+** and **124-** are proximate to waveguide arm **132**. The electrical signal carried by differential electrode **124-** modulates the optical signal in waveguide arm **134** in regions where differential electrode **124-** is proximate to arm **134** of waveguide **130**. For example, the location of portions of waveguide **130** may be further from electrodes **120** than appears in the plan view of FIG. **1A**. For example, portions of waveguide **130** may be separated by a distance out of the plane of the page for FIG. **1A** (e.g., portions of waveguide **130** may be buried deeper in the device **100** or portions of electrode trio **124** may be further from substrate structure **101**). The modulated optical signal for each arm **132** and **134** of a waveguide **130** may be recombined and then output.

[0036] Electro-optic device **100** may have engineered chirp. Chirp refers to frequency components generated through modulation that do not contribute to amplitude modulation. For example, pure phase modulation generates sidebands without adding amplitude modulation. In intensity modulations such as Mach Zehnder or Michelson interferometers, chirp can be generated by performing different modulation on the two modulator arms that do not cancel each other in the beam splitter of the interferometer. While chirp on its own does not generate amplitude modulation, by propagating through a dispersive medium the relative spectral phase changes. This phase change in turn can lead to amplitude modulation or distortion of modulated waveforms. Therefore the ability to control chirp in electro-optic modulation is desirable. Electro-optic device **100** includes waveguide arms **132** and **134** and electrode trio **124** that are configured to provide such controlled chirp.

[0037] For electro-optic materials (e.g., TFLC materials), the electric field may produce refractive index changes through the electro-optic effect. For TFLC materials, the modulation depends upon the magnitude and direction of the electric field. In electro-optic device **100**, the electric fields generated by differential electrodes **124+** and **124-** and by electrode **124-** and ground electrode **124G** are in opposite direction. In addition, the electrodes **124+**, **124-**, and **124G** may be configured to provide different amounts (i.e. different magnitudes) of modulation (in addition to different directions of modulation) of the optical signals in waveguide arms **132** and **134**. Consequently, a controlled, or engineered, chirp may be provided in electro-optic device **100**.

[0038] More specifically, waveguide arm **132** is between differential electrodes **124+** and **124-**. Differential electrodes **124+** and **124-** carry electrical signals of opposite polarity (e.g. complimentary electrical signals). The electrical signals carried by electrodes **124+** and **124-** generate an electric field between differential electrodes **124+** and **124-**. The electric field is in the region of waveguide arm **132** and modulates the index of refraction of waveguide arm **132**. Thus, the optical signal carried by waveguide arm **132** is modulated. Differential electrodes **124+** and **124-** carrying complementary electrical signals form a push-pull configuration for the intensity modulation. As a result, the required voltage to drive waveguide arm **132** to have the desired modulation (e.g. the desired V- π) may be reduced.

[0039] In contrast, waveguide arm **134** is between negative differential electrode **124-** and ground electrode **124G**. The electrical signal carried by negative differential electrode **124-** generates an electric field between electrodes **124-** and **124G**. This electric field modulates the index of refraction of waveguide arm **134**. Consequently, the optical signal carried by waveguide arm **134** is

modulated. However, the electric field between electrodes **124-** and **124G** is in the opposite direction to the electric field between differential electrodes **124+** and **124-**. Moreover, differential electrode **124-** and ground **124G** do not form a push-pull configuration. Consequently, the voltage required to drive waveguide arm **134** may not be reduced.

[0040] The modulation provided to the optical signal in arm **132** or **134** of waveguide **130** depends upon the electric field and the length of the waveguide arms **132** and/or **134** along which the electric field is applied. The electric field applied depends upon the electrical signal through differential electrodes **124+** and **124-**, the distances between electrodes **124+** and **124-** and between electrodes **124-** and **124G**. For example, if the gap between electrodes **124+** and **124-** is the same as the gap between electrodes **124-** and **124G** and electrical signals having the same magnitude and opposite polarity (i.e. complementary, differential signals) are driven through electrodes **124+** and **124-**, then the modulation across waveguide arm **132** may be twice that of the modulation across waveguide arm **134** and in the opposite direction. Thus, by controlling the gap between electrodes **124+** and **124-**, the gap between electrodes **124-** and **124G**, the electrical signal driven through electrodes **124+** and **124-**, and the length along which the electrode trio **124** is in proximity to waveguide arms **132** and **134** (e.g. the $V \cdot \pi \cdot L$ for the waveguide arms **132** and **134**) the chirp introduced into waveguide **130** may be controlled.

[0041] For example, suppose differential electrodes **124+** and **124-** carry a particular electrode signal and are separated by a gap (i.e. provide an electric field for waveguide arm **132**) that generates a modulation having magnitude M_1 in waveguide arm **132**. Electrodes **124-** and **124G** are configured such that the electric field for waveguide arm **134** generates a modulation having magnitude M_2 in waveguide arm **132**. Electro-optic device **100** thus has a chirp parameter, C , given by $C = (M_1 - M_2) / (M_1 + M_2)$. In some embodiments, electro-optic device **100** is configured to have a chirp parameter of at least 0.1 and not greater than 0.9 or not greater than -0.1 and at least -0.9 (i.e., $-0.9 \leq C \leq -0.1$ or $0.1 \leq C \leq 0.9$). Thus, the chirp, or chirp parameter, may be engineered for electro-optic device **100**.

[0042] FIG. 1C depicts electro-optic device **100'**. Electro-optic device **100'** includes waveguide **130'** and electrodes **120** including electrode trio **124** and ground electrode **126**. Electrode trio **124** includes differential electrodes **124+** and **124-** and ground electrode **124G** that are analogous to differential electrodes **124+** and **124-** and ground electrode **124G** of electro-optic device **100**. Waveguide **130'** is analogous to waveguide **130**. Thus, waveguide arms **132'** and **134'** are analogous to waveguide arms **132** and **134**. However, in electro-optic device **100'**, arm **132'** is between ground electrode **126** and positive differential electrode **124+**, while waveguide arm **134'** is between differential electrodes **124+** and **124-**. Similarly, electrode **124+** is between waveguide arms **132** and **134**. In some embodiments, two electrodes carrying the same signal may be used instead of a single electrode **124+**.

[0043] Electro-optic device **100'** operates in an analogous manner to electro-optic device **100**. However, waveguide arm **134'** is exposed to the push-pull configuration, while waveguide arm **132'** is not. Thus, the modulation, and chirp, for electro-optic device **100'** is in the opposite direction as that of electro-optic device **100**. For example, for equally spaced electrodes **124+**, **124-**, and **126**, the electric field applied to waveguide arm **134'** is approximately twice the magnitude and in the opposite direction of the electric field applied to waveguide arm **132'**. By configuring the gaps between electrodes **120**, the signal through differential electrodes **124+** and **124-**, and the lengths of waveguide arms **132'** and **134'** along which the electric field is applied, the modulation to each arm **132'** and **134'** is controlled. A controlled chirp having an opposite direction may be provided by electro-optic device **100'**.

[0044] Thus, electro-optic devices **100** and **100'** may provide an engineered chirp to waveguides **130** and **130'**. Stated differently, electro-optic devices **100** and **100'** may have tailored chirp parameters. Controlling the chirp introduced into waveguides **130** and **130'** may improve performance. For example, chirp introduced in other components may be accounted for. In one

such example, dispersion in optical fibers to which waveguide **130** or **130'** are coupled may be mitigated by introducing in waveguide **130** and **130'** chirp opposite to the dispersion. As a result, the optical signal may be better maintained and longer transmission distances may be achieved. Furthermore, the chirp induced using electro-optic devices **100** and **100'** may not change the loss in waveguides **130** and/or **130'**. Consequently, the technique for inducing chirp may not change the extinction ratio of the Mac-Zehnder interferometer configuration of electro-optic devices **100** and/or **100'**. The extinction ratio of electro-optic devices **100** and/or **100'** may be maintained to be at least 10 dB, at least 15 dB, at least 20 dB or at least 25 dB in various embodiments. In some embodiments, the extinction ratio may be less than 100 dB. As a result, the chirp may be engineered without adversely affecting the extinction ratio of electro-optic devices **100** and/or **100'**. Consequently, performance of a system utilizing electro-optic devices **100** and/or **100'** may be improved.

[0045] FIG. 2 depicts a plan view of an embodiment of electro-optic device **200** having engineered chirp. For clarity, not all components are shown and FIG. 2 is not to scale. FIG. 2 may be considered to show portions of electro-optic device **200**. For example, drivers, voltage sources, the source of the optical signal, and other portions of such a system are not shown. Electro-optic device **200** is analogous to electro-optic device **100** and, therefore, functions in a similar manner.

[0046] Electro-optic device **200** includes electrodes **220** and waveguide **230**, that are analogous to electrodes **120** and waveguide **130**. Thus, waveguide **230** includes or consists of TFLC materials such as TFLN and/or TFLT. Electro-optic device **200** may also include cladding and a substrate structure analogous to cladding **140** and substrate structure **101**. Waveguide **230** includes arms **232** and **234** that are analogous to arms **132** and **134** of waveguide **100**. Similarly, electrodes **220** includes ground electrode **226** and electrode trio **124** that are analogous to ground electrode **126** and electrode trio **124**. Electrode trio **224** includes differential electrodes **224+** and **224-** and ground electrode **224G** that are analogous to differential electrodes **124+** and **124-** and ground electrode **124G**. Thus, differential electrodes **224+** and **224-** may carry signals that are opposite in polarity. The frequency of the electrical signals carried by differential electrodes **224+** and **224-** may be high, for example in the 100 kHz through 100 GHz (or higher) range. Ground electrode **224G** may also have a slow varying potential or a constant bias (zero or nonzero) voltage. Waveguide arm **232** is between differential electrodes **224+** and **224-**, while waveguide arm **234** is between signal electrode **224-** and ground electrode **224G**.

[0047] Although waveguide arm **232** is between differential electrodes **224-** and **224+**, positive electrode **224+** is between waveguide arms **232** and **234**. This is different from the locations of the positive electrode **124+** and negative electrode **124-** in electro-optic device **100**. However, waveguide arm **232** and electrodes **224+** and **224-** are still in a push-pull configuration. Further, waveguide arm **234** is between electrode **224+** and ground electrode **224G**. Thus, optical signals through waveguide arms **232** and **234** are modulated in an analogous manner to the optical signal through waveguide arms **132** and **134**. However, the electric fields may be in the opposite direction.

[0048] Also explicitly labeled are gaps **252** and **254** having widths d_1 and d_2 , respectively. Gap **252** is between differential electrodes **224+** and **224-** and is traversed by waveguide arm **232**. Gap **254** is between differential electrode **224+** and ground electrode **224G** and is traversed by waveguide arm **234**. In the embodiment shown, gap **252** between differential electrodes **224+** and **224-** is wider than gap **254** between electrodes **224+** and **224G** (i.e. $d_1 > d_2$). However, the gap length(s) d_1 and/or d_2 may be different. For example, gap **252** may be the same size as gap **254** ($d_1 = d_2$), smaller than gap **254** ($d_1 < d_2$), or a smaller amount larger than gap **254** ($d_1 > d_2$, but by a smaller amount). Gaps **252** and **254** are each not more than twenty micrometers wide in some embodiments. In some embodiments, each of gaps **252** and **254** may have length of at least 2 micrometers (d_1 , d_2 each at least 2 micrometers). In some embodiments, gap **252** and/or gap **254** has a length of at least 2.5 micrometers (d_1 and/or d_2 at least 2.5 micrometers). Each gap **252** and

254 may have a width that is not more than ten micrometers ($d1$, $d2$ each not more than 10 micrometers). In some embodiments, gap **252** and/or gap **254** is not more than six micrometers ($d1$ and/or $d2$ not more than 6 micrometers).

[0049] Electro-optic device **200** operates in an analogous manner to electro-optic devices **100** and/or **100'**. Thus, a controlled chirp may be introduced. The electric field at waveguide arms **232** and **234** may be proportional to $V \cdot \pi \cdot L$, where V is the magnitude of the electrode signals and L is the length along which the electrode trio **234** is proximate to waveguide arms **232** and **324**. By controlling gaps **252** and **254** and/or $V \cdot \pi \cdot L$, the modulation and thus the chirp can be configured. In some embodiments, gap **254** between the differential electrode **224+** and ground may be desired to be at a minimum without inducing significant optical loss to maintain highest electro-optic modulation efficiency. In some embodiments, gap **254** between signal electrode **224+** and ground electrode **224G** may be much wider to achieve more extreme chirp at the cost of electro-optic modulation efficiency. One or both of gaps **252** and **254** may be set to have widths in the above (or other) ranges to change the electric field at waveguide arms **232** and **234** and, therefore, the chirp. The electric field at waveguide arms **232** and **234** is also proportional to $V \cdot \pi \cdot L$, where V is the magnitude of the electrode signals and L is the length along which the electrode trio **234** is proximate to waveguide arms **232** and **324**. Thus, by controlling gaps **252** and **254** and/or $V \cdot \pi \cdot L$, the modulation and thus the chirp can be configured.

[0050] For example, if gaps **252** and **254** are the same size ($d1=d2$), then the electric field experienced by waveguide arm **232** is approximately twice the electric field experienced by waveguide **234** for a differential signal driven through differential electrodes **224+** and **224-**. Thus, a chirp is introduced. In this example, $d1$ and $d2$ are set such that $M1$ is twice as large as $M2$. In such embodiments, electro-optic device **200** has chirp parameter of $C=1/3$. If a smaller magnitude chirp is desired, then during fabrication, gap **252** is increased in size and/or gap **254** is decreased in size. Gaps **252** and/or **254** may be configured such that electro-optic device **200** has a chirp parameter of at least 0.1 and not greater than 0.9 or not greater than -0.1 and at least -0.9, where $C=(M1-M2)/(M1+M2)$ ($-0.9 \leq C \leq -0.1$ or $0.1 \leq C \leq 0.9$). For example, in one embodiment, gap **252** may have a width of four micrometers, while gap **254** has a width of three micrometers. In another example, if gap **252** between the signal (the differential) electrodes **124+** and **124-** is twice gap **252** between signal electrode **224+** and ground **224G**, then approximately zero chirp may be introduced. Depending upon the relative widths of the gaps, positive or negative chirps may be introduced. If gap **252** between the differential (i.e. signal) electrodes **224+** and **224-** is greater than twice gap **254** between signal electrode **224+** and ground electrode **224G**, the chirp introduced has one sign. If gap **252** between differential electrodes **224+** and **224-** is less than twice gap **254** between signal electrode **224+** and ground electrode **224G**, the chirp introduced has the opposite sign. In some embodiments, the electrical signal applied to differential electrodes **224+** and **224-** is 0.8V-1V differential peak-to-peak. In some embodiments, the signal applied may be lower than 0.8 volts peak-to-peak. In some embodiments, the signal is greater than 1V. For example, the signal may be 1.5V differential peak-to-peak. However, the signal may be desired to be not more than 3.5V differential peak-to-peak.

[0051] Thus, gaps **252** and **254** between the electrodes **224-** and **224+** and electrodes **224+** and **224G** may be configured to control chirp independently from the signal through differential electrodes **224+** and **224-**. In some embodiments the width selected for gaps **252** and **243** may change the sign of the chirp, but not the magnitude. In some embodiments, the magnitude of the chirp may also be controlled. Consequently, electro-optic device **200** may share the benefits of electro-optic devices **100** and/or **100'**. For example, the engineered chirp introduced by electro-optic device **200** may be used to account for chirp and/or dispersion introduced in other components. In one example, dispersion due to optical fibers (not shown) may be mitigated. As a result, the optical signal may be better maintained and longer transmission distances may be achieved. Further, the introduction of controlled chirp may not adversely affect the extinction ratio

of electro-optic device **200**. Consequently, performance of a system utilizing electro-optic device **200** may be improved.

[0052] FIG. **3** depicts a plan view of an embodiment of electro-optic device **300** having engineered chirp. For clarity, not all components are shown and FIG. **3** is not to scale. FIG. **3** may be considered to show portions of electro-optic device **300**. For example, drivers, voltage sources, the source of the optical signal, and other portions of such a system are not shown. Electro-optic device **300** includes two devices, each of which is analogous to device(s) **100**, **100'**, and/or **200**. Thus, electro-optic device **300** includes waveguides **330-1** and **330-2** (collectively or generically waveguide(s) **330** or waveguide **330-i**). Each waveguide **330-i** (where *i* is 1 or 2) includes arms **332-i** and **334-i** (collectively or generically arm(s) **332** and **334**). Electro-optic device also includes electrodes **320-i** (collectively or generically electrodes **320**) including ground electrode **326-i** (collectively or generically ground electrode **326**) and electrode trio **324-i** (collectively or generically electrode trio **324**), where *i* is 1 or 2. Each trio **324-i** includes differential electrodes **324+-i** and **324--i** (collectively or generically differential electrodes **324+** and **324-**) and ground electrodes **324G-i** (collectively or generically ground electrode **324G**), where *i* is 1 or 2. Waveguide arm **332** is between differential electrodes **324+** and **324-**. Waveguide arms **334** are between differential electrodes **324+** and ground electrode **324G**.

[0053] Electrodes **320** and waveguides **330** are analogous to electrodes **120**, **220** and **320** and waveguides **130**, **130'**, and **230**. Thus, waveguides **330** includes or consists of TFLC materials such as TFLN and/or TFLT. Electro-optic device **300** may also include cladding and a substrate structure analogous to cladding **140** and substrate structure **101**. Chirp for each waveguide **330** may be controlled in a manner analogous to that discussed for electro-optic devices **100**, **100'**, and/or **200**. For example, the gaps between electrodes **320**, the signal applied to differential electrodes **324+** and **324-**, and the lengths of electrodes **320** proximate to waveguide arms **332** and **334** can be engineered to provide the desired chirp.

[0054] More specifically, electrodes **320-1** and waveguide **330-1** are configured in an analogous manner to electrodes **220** and waveguide **230**. Gaps **352** and **354** are analogous to gaps **252** and **254**. Thus, waveguide **330-1** has an engineered chirp controlled by the widths of gaps **352** and **354**. However, the reduction in the electric field between differential electrodes **324+-1** and **324--1** may affect the efficiency of modulation for waveguide **330-1**. $V\text{-}\pi^*$ for waveguide **330-1** is less it would be if gap **352** were narrower (e.g. the same as the gaps for electrodes **320-2**). To address this change in efficiency, the lengths of electrodes may be configured. Electrodes **324+-2** and **324--2**, **324G-2**, and **326-2** having smaller gaps have a reduced length L_1 , while electrodes **324+-2** and **324--2**, **324G-2**, and **326-2** with larger gap **352** have lengths $L > L_1$ in the region of waveguides **330**. By reducing the length of differential electrodes **324+-2** and **324--2** to L_1 , $V\text{-}\pi^*L$ for waveguide **330-2** has been reduced. In some embodiments, $V\text{-}\pi^*L$ for waveguides **320-1** and **320-2** may be the same. Thus, gaps **352** and **354** may be used to engineer chirp and electrode lengths may be used to control $V\text{-}\pi$.

[0055] Electro-optic device **300** may share the benefits of electro-optic devices **100**, **100'** and/or **200**. For example, the engineered chirp introduced by electro-optic device **300** may be used to account for chirp and/or dispersion introduced in other components. In one example, dispersion due to optical fibers (not shown) may be mitigated. As a result, the optical signal may be better maintained and longer transmission distances may be achieved. Further, $V\text{-}\pi$ may also be controlled. In addition, the introduction of controlled chirp may not adversely affect the extinction ratio of electro-optic device **300**. Consequently, performance of a system utilizing electro-optic device **300** may be improved.

[0056] Although described in the context of one or two waveguides (or single pairs of waveguides/waveguide arms), multiple gaps, multiple waveguide pairs, and multiple electrode pairs may be used in some embodiments. For example, FIGS. **4-6** depict embodiments of electro-optic devices **400**, **500**, and **600** which combine aspects of multiple devices analogous to electro-

optic devices **100**, **100'**, **200**, and/or **300**. Other combinations that are not shown are also possible. [0057] FIG. **4** depicts an embodiment of electro-optic device **400** having engineered chirp. For clarity, not all components are shown and FIG. **4** is not to scale. For simplicity, drivers, voltage sources, the source of the optical signal, and other portions of such a system are not shown. Electro-optic device **400** includes four devices, each of which is analogous to device(s) **100**, **100'**, **200**, and/or **300**. Thus, electro-optic device **400** includes waveguides **430-1**, **430-2**, **430-3**, and **430-4** (collectively or generically waveguide(s) **430** or waveguide **430-i**). Each waveguide **430-i** (where *i* is 1, 2, 3, or 4) includes arms **432-i** and **434-i** (collectively or generically arm(s) **432** and **434**). Electro-optic device also includes electrodes **420-i** (collectively or generically electrodes **420**) including ground electrode **426-i** (collectively or generically ground electrode **426**) and electrode trio **424-i** (collectively or generically electrode trio **424**), where *i* is 1, 2, 3, or 4. Each trio **424-i** includes differential electrodes **424+-i** and **424--i** (collectively or generically differential electrodes **424+** and **424-**) and ground electrodes **424G-i** (collectively or generically ground electrode **424G**), where *i* is 1, 2, 3, or 4. Waveguide arm **432** is between differential electrodes **424+** and **424-**. Waveguide arms **434** are between differential electrodes **424-** and ground electrode **424G**. Electrodes **420** and waveguides **430** are, therefore, analogous to electrodes **120**, **220** and **320** and waveguides **130**, **130'**, **230**, and **330**. Thus, waveguides **430** includes or consists of TFLC materials such as TFLN and/or TFLT. Electro-optic device **400** may also include cladding and a substrate structure analogous to cladding **140** and substrate structure **101**. Chirp for each waveguide **430** may be controlled in the manner analogous to that discussed for electro-optic devices **100**, **100'**, **200**, and/or **300**. For example, the gaps between electrodes **420**, the signal applied to differential electrodes **424+** and **424-**, and the lengths of electrodes **420** proximate to waveguide arms **432** and **434** can be engineered to provide the desired chirp.

[0058] In some embodiments, electro-optic device **400** is a four channel transmitter device that may use optical signals having four different wavelengths (one wavelength for each waveguide **430**). Electro-optic device **400** may thus be considered to include four modulators (one for each waveguide **430**). Electro-optic device **400** may improve transmission of optical signals. The wavelengths of the optical signals may be chosen such that when propagating through optical fibers, the optical signals experience positive (normal), negative (anomalous), or zero dispersion. For example, optical fiber(s) (not shown) coupled with electro-optic device **400** may have zero dispersion for optical signals having a 1311 nanometer wavelength, nonzero dispersion (e.g. negative) for optical signals having wavelengths above 1311 nanometers, and nonzero dispersion having the opposite sign (e.g. positive) for wavelengths below 1311 nanometers. Non-zero dispersion adversely affects data transmission quality, particularly for transmission over a long distance. To address this issue, controlled chirp may be introduced for one or more waveguides **430**. The chirp introduced for each waveguide **430** may be determined by the wavelength of the optical signal carried by each waveguide **430** and the distance targeted. The controlled chirp provided by electro-optic device **400** counters some or all of the dispersion introduced by propagation in a corresponding optical fiber over a distance. Thus, electro-optic device **400** improves transmission of the four channels transmitted via optical device **400**.

[0059] The effect of the controlled chirp may be understood as follows. Spectral chirp can be described in ps/nm, meaning a certain ps of chirp per nm bandwidth. The fiber dispersion parameter for a system describes how much chirp is added per kilometer of propagation, given in ps/nm/km. For example, assume that for an optical signal having a wavelength of 1350 nm, the fiber dispersion parameter is -3 ps/nm/km and the optical signal propagates for two kilometers. The chirp added by dispersion in the optical fiber is -6 ps/nm. Suppose the optical fiber is coupled with waveguide **430-2**, which carries an optical signal having a wavelength of 1350 nm. In order to compensate for the fiber dispersion the modulator for waveguide **430-2** in electro-optic device **400** may produce $+6$ ps/nm spectral chirp for the optical signal. In such a case, the controlled chirp introduced by electro-optic device **400** and the dispersion in the optical fiber add up such that the

optical signal at the destination has no chirp and no distortion. In practice the chirp introduced to control dispersion may be slightly more or less than +6 ps/nm. The chirp introduced to the optical signals in remaining waveguides **430-1**, **430-3**, and **430-4** may be similarly controlled. Thus, the use of electro-optic device **400** may improve transmission of optical signals.

[0060] An analogous technique can extend this chirp/dispersion control to a higher or lower number of channels. For more wavelengths (i.e. more channels), each channel has slightly different chirp and therefore uses different compensation introduced by optical device **400**. For a SMF28e fiber, for example, typical distances of propagation (e.g. between links) are at least 1 km to not more than 50 km. In such a case, the dispersion may be between -5 ps/nm/km and +5 ps/nm/km. The chirp introduced may be desired to be between -5 ps/nm and +5 ps/nm (e.g. for a 1 km propagation length) to -250 ps/nm and +250 ps/nm (e.g. for a 50 km propagation distance). In some embodiments, the sign of the chirp introduced may be determined by the gaps selected for waveguide arms **432** and **434** (e.g., the gap between differential electrodes **424+** and **424-**, the gap between the positive electrode **424+** and ground electrode **426**, and the gap between negative electrode **424-** and ground **424G**). Thus, the chirp for each waveguide **430** (and thus each channel) may be independently controlled to improve transmission of optical signals. Electro-optic device **400** may thus share the benefits of electro-optic device(s) **100**, **200**, and/or **300**. For example, each channel may have independently controlled chirp and the introduction of controlled chirp may not adversely affect the extinction ratio of the channel(s) of electro-optic device **400**. Optical device **400** may thus introduce controlled chirp in each channel that compensates for dispersion in optical fibers coupled with waveguides **430**. As a result, the optical signal may be better maintained and longer transmission distances may be achieved. Consequently, performance of an optical network using electro-optic device **400** may be improved.

[0061] FIG. 5 depicts an embodiment of an electro-optic device having engineered chirp. For clarity, not all components are shown and FIG. 5 is not to scale. For simplicity, drivers, voltage sources, the source of the optical signal, and other portions of such a system are not shown. Electro-optic device **500** is analogous to electro-optic device **400**. Thus, four devices, each of which is analogous to device(s) **100**, **100'**, **200**, and/or **300**, are included in electro-optic device **500**. Another number of devices (i.e. channels) may be included in other embodiments.

[0062] Electro-optic device **500** includes waveguides **530-1**, **530-2**, **530-3**, and **530-4** (collectively or generically waveguide(s) **530** or waveguide **530-i**). Each waveguide **530-i** (where i is 1, 2, 3, or 5) includes arms **532-i** and **534-i** (collectively or generically arm(s) **532** and **534**). Electro-optic device also includes electrodes **520-i** (collectively or generically electrodes **520**) including ground electrode **526-i** (collectively or generically ground electrode **526**) and electrode trio **524-i** (collectively or generically electrode trio **524**), where i is 1, 2, 3, or 5. Each trio **524-i** includes differential electrodes **524+-i** and **524--i** (collectively or generically differential electrodes **524+** and **524-**) and ground electrodes **524G-i** (collectively or generically ground electrode **524G**), where i is 1, 2, 3, or 5. Waveguide arm **532** is between differential electrodes **524+** and **524-**. Waveguide arms **534** are between differential electrodes **524-** and ground electrode **524G**. Electrodes **520** and waveguides **530** are, therefore, analogous to electrodes **120**, **220**, **320** and **420** and waveguides **130**, **130'**, **230**, **330**, and **430**. Thus, waveguides **530** includes or consists of TFLC materials such as TFLN and/or TFLT. Electro-optic device **500** may also include cladding and a substrate structure analogous to cladding **140** and substrate structure **101**. Chirp for each waveguide **530** may be controlled in a manner analogous to that discussed for electro-optic devices **100**, **100'**, **200**, **300**, and/or **400**. For example, the gaps between electrodes **520**, the signal applied to differential electrodes **524+** and **524-**, and the lengths of electrodes **520** proximate to waveguide arms **532** and **534** can be engineered to provide the desired chirp.

[0063] In electro-optic device, the locations of waveguide arms **532** and **534** as well as the gaps between electrodes **520** may be selected to provide the desired chirp for each channel. In electro-optic device **500**, gap **552** between differential electrodes **524+-2** and **524--2** and gap **554** between

differential electrodes **524+3** and **524-3** are controlled to be different from (e.g. wider than) the other gaps between the remaining electrodes of electro-optic device **500**. For waveguide **530-1**, the gaps between electrodes **520-1** are the same. In addition, waveguide arm **532-1** is between differential electrodes **524+-1** and **524--1**, while waveguide arm **534** is between differential electrode **524--1** and ground electrode **524G-1**. Thus, the modulation of the optical signal for waveguide arm **532-1** is approximately twice the modulation for the optical signal for waveguide arm **534-1**.

[0064] For waveguide **530-2**, gap **552** between differential electrodes **524+-2** and **524--2** is larger than the gap between differential electrode **524--2** and ground electrode **524G-2**. This changes the difference in modulation between waveguide arms **532-2** and **534-2**. As long as gap **552** is not more than twice the remaining gaps for electrodes **520-2**, the difference in modulation between waveguide arms **532-2** and **534-2** (and thus the chirp for waveguide **530-2**) is reduced from that of waveguide **530-1**. The optical signal in waveguide arm **532-2** still experiences greater modulation than the optical signal in waveguide arm **534-2**.

[0065] For waveguide **530-3**, waveguide arm **532-3** is between ground electrode **526-3** and positive differential electrode **524+-3**. Waveguide arm **534-3** is between differential electrodes **524+-3** and **524--3**. In addition, gap **554** is wider than the remaining gaps between electrodes **520-3**. This changes the difference in modulation between waveguide arms **532-3** and **534-3**. As long as gap **554** is not more than twice the remaining gaps for electrodes **520-3**, the difference in modulation between waveguide arms **532-3** and **534-3** (and thus the chirp for waveguide **530-3**) is reduced in magnitude from that of waveguide **530-1**. However, because of gap **554** and the placement of waveguide arms **532-3** and **534-3** between electrodes **520-3**, the chirp has the opposite sign from the chirp introduced to the optical signal carried by waveguide **530-1**.

[0066] For waveguide **530-4**, waveguide arm **532-4** is between ground electrode **526-4** and positive differential electrode **524+-4**. Waveguide arm **534-4** is between differential electrodes **524+-4** and **524--4**. In addition, the gaps between electrodes **520-4** are all substantially the same. This changes the difference in modulation between waveguide arms **532-3** and **534-3**. The modulation between waveguide arms **532-4** and **534-4** (and thus the chirp for waveguide **530-4**) is substantially the same in magnitude as that of waveguide **530-1**. However, because of the placement of waveguide arms **532-4** and **534-4** between electrodes **520-4**, the chirp has the opposite sign from the chirp introduced to the optical signal carried by waveguide **530-1**.

[0067] Thus, each waveguide **530** has a chirp that is individually controlled in both strength (e.g. magnitude) and sign by engineering the size of gaps (e.g. gaps **552** and **554**) and placement of waveguides **530** between electrodes **520**. In some embodiments, instead of changing the placement of waveguides **530**, the positions of electrodes **524+-i** and **524--i** may be switched. In some embodiments, nonzero chirp is introduced to each waveguide **530**. In some embodiments, electro-optic device **500** is configured such that not all waveguides **530** have a controlled chirp introduced. For some applications, only waveguides **530** carrying optical signals of very chirped wavelength may be adjusted to have a compensating chirp. In some embodiments, the sign of the chirp introduced is changed by choosing the gaps, the magnitude by changing the relative gap values. Because each channel may carry a different wavelength, each channel experiences different chirp due to fiber propagation. Consequently, it may be advantageous to adjust the chirp separately for each waveguide **530**.

[0068] Optical device **500** may introduce controlled chirp in each channel that compensates for dispersion in optical fibers and/or other components coupled with waveguides **530**. As a result, the optical signal may be better maintained and longer transmission distances may be achieved. Electro-optic device **500** may thus share the benefits of electro-optic device(s) **100**, **200**, and/or **300**. For example, each channel may have independently controlled chirp and the introduction of controlled chirp may not adversely affect the extinction ratio of the channel(s) of electro-optic device **500**. Consequently, performance of an optical network using electro-optic device **500** may

be improved.

[0069] FIG. 6 depicts an embodiment of an electro-optic device having engineered chirp. For clarity, not all components are shown and FIG. 6 is not to scale. For simplicity, drivers, voltage sources, the source of the optical signal, and other portions of such a system are not shown. Electro-optic device **600** is analogous to electro-optic device(s) **400** and/or **500**. Thus, four devices, each of which is analogous to device(s) **100**, **100'**, **200**, and/or **300**, are included in electro-optic device **600**. Another number of devices (i.e. channels) may be included in other embodiments. [0070] Electro-optic device **600** includes waveguides **630-1**, **630-2**, **630-3**, and **630-4** (collectively or generically waveguide(s) **630** or waveguide **630-i**). Each waveguide **630-i** (where *i* is 1, 2, 3, or 6) includes arms **632-i** and **634-i** (collectively or generically arm(s) **632** and **634**). Electro-optic device also includes electrodes **620-i** (collectively or generically electrodes **620**) including ground electrode **626-i** (collectively or generically ground electrode **626**) and electrode trio **624-i** (collectively or generically electrode trio **624**), where *i* is 1, 2, 3, or 6. Each trio **624-i** includes differential electrodes **624+-i** and **624--i** (collectively or generically differential electrodes **624+** and **624-**) and ground electrodes **624G-i** (collectively or generically ground electrode **624G**), where *i* is 1, 2, 3, or 6. Waveguide arm **632** is between differential electrodes **624+** and **624-**. Waveguide arms **634** are between differential electrodes **624-** and ground electrode **624G**. Electrodes **620** and waveguides **630** are, therefore, analogous to electrodes **120**, **220**, **320** and **420** and waveguides **130**, **130'**, **230**, **330**, and **430**. Thus, waveguides **630** includes or consists of TFLC materials such as TFLN and/or TFLT. Electro-optic device **600** may also include cladding and a substrate structure analogous to cladding **140** and substrate structure **101**. Chirp for each waveguide **630** may be controlled in the manner analogous to that discussed for electro-optic devices **100**, **100'**, **200**, **300**, and/or **400**. For example, the gaps between electrodes **620**, the signal applied to differential electrodes **624+** and **624-**, and the lengths of electrodes **620** proximate to waveguide arms **632** and **634** can be engineered to provide the desired chirp. In electro-optic device, the locations of waveguide arms **632** and **634** as well as the gaps between electrodes **620** may be selected to analogous to that of waveguide arms **532** and **534** as well as gaps between electrodes **520**. Thus, the chirp for waveguides **630-2** and **630-3** is controlled at least in part through the gaps between electrodes **620**.

[0071] Engineering the chirp through the gaps between electrodes **620** can cause a reduction in the electric field provided to the waveguide arms **632** or **634**. This reduction in electric field affects the V- π of the modulator. More specifically, the change in the gap affects the field at the waveguide, and thus the modulation. V- π may, therefore, change. In some embodiments the difference in V- π may be compensated for by driving the channels with different voltage swings. In some embodiments, the same voltage swing may be used for each channel (e.g. each waveguide **630**). Electro-optic device **600** is one such device. Because electrical signals in electrodes **624+** and **624-** are the same for each waveguide **630**, different lengths and/or design of electrodes **624+** and **624-** may be used to achieve a more uniform V- π and bandwidth response. In electro-optic device **600**, electrodes **624+-1**, **624--1**, **624+-4**, and **624--4** are shorter than electrodes **624+-2**, **624--2**, **624+-3**, and **624--3**. Electrodes **620-1** are more efficient because the uniform gaps between electrodes **620-1** are smaller. Electrodes **620-4** are more efficient for similar reasons. Electrodes **620-2** and **620-3** have wider gaps **652** and **654**. Therefore, electrodes **620-2** and **620-3** (length *l*) are longer to achieve the same driving voltage benchmarks as electrodes **620-1** and **620-4**. To manage V- π , therefore, electrodes **620-1** and **620-4** (length *l*₁) are shorter than electrodes **620-2** and **620-3** (length *l*). In some embodiments, the difference in length may be a few millimeters.

[0072] Electrodes **620-2** and **620-3** may suffer greater microwave loss at high frequencies due to propagation loss. This can be offset by designing the electrodes to be different. For example, the shorter electrodes **620-1** and **620-4** may use regular uniform channel electrode(s), whereas the longer electrodes **620-2** and **620-3** may use different electrode(s), for example electrodes with extensions, to achieve a similar high speed response in addition to V- π as the shorter electrodes

620-1 and **620-4**. Thus, V-pi and chirp can also be independently controlled together by changing the relative gaps between the signal and ground electrodes and the lengths of electrodes.

[0073] Optical device **600** may introduce controlled chirp in each channel that compensates for dispersion in optical fibers and/or other components coupled with waveguides **630**. As a result, the optical signal may be better maintained and longer transmission distances may be achieved. Electro-optic device **600** may thus share the benefits of electro-optic device(s) **100**, **200**, and/or **300**. For example, each channel may have independently controlled chirp and the introduction of controlled chirp may not adversely affect the extinction ratio of the channel(s) of electro-optic device **600**. Consequently, performance of an optical network using electro-optic device **600** may be improved.

[0074] FIG. 7 is a flow chart depicting an embodiment of method **700** for using an electro-optic device having engineered chirp. Thus, method **700** may be used to modulate optical signal(s). Method **700** is described in the context of processes that may have sub-processes. Although described in a particular order, another order not inconsistent with the description herein may be utilized. Further, although described in the context of a single optical signal, method **700** may be extended to multiple optical signals and multiple differential signals.

[0075] An optical signal is provided to a first waveguide, at **702**. A second optical signal is provided to a second waveguide, at **704**. In some embodiments, the first and second waveguides are arms of a waveguide (e.g. the arms of a Mach-Zehnder interferometer). Thus, **702** and **704** may include providing a single optical signal to a waveguide including splitter(s). The waveguide carries the signal and, because of the presence of the splitter(s), divides the signal between at least two arms of the waveguide. Thus, **702** and **704** may be carried out by providing different optical signals or by splitting an optical signal.

[0076] Electrode signals are driven through electrodes that correspond to the first and second waveguides, at **706**. The electrodes may form a first differential electrode pair proximate to a portion of the first waveguide and a second differential electrode pair proximate to the second waveguides. The electrode signals in first differential electrode pair and the second differential electrode pair are configured to provide a first modulation to the first optical signal and a second modulation to the second optical signal. The first modulation is different from the second modulation. For example, in some embodiments, the electrode signals themselves may differ in a manner that results in different modulations in the first and second waveguides. In some embodiments, the electrodes and waveguides are configured such that the first and second modulations are different. Thus, the electro-optic device has an engineered chirp.

[0077] For example, method **700** may be used with electro-optic device **100**. In such an embodiment, **702** and **704** are performed by providing an optical signal to waveguide **130**, which splits the optical signal and provides a portion of the optical signal to each of waveguide arms **132** and **134**. Thus, waveguide arm **132** may be the first waveguide, while waveguide arm **134** is the second waveguide. At **706**, an electrode signal is driven through electrodes **124+** and **124-**. In such an embodiment, the first differential electrode pair that modulates the signal in waveguide **132** are electrodes **124+** and **124-**. The second differential electrode pair that modulates the signal in waveguide **134** are negative electrode **124-** and ground electrode **124G**. Because of the configuration of electrodes **124+**, **124-**, and **124G** and waveguide **130**, the electrode signals drive through electrodes **124+** and **124-** result in different modulations in waveguide arms **132** and **134**. Consequently, a controlled chirp is intentionally provided to waveguide **130**. Method **700** may be used in an analogous manner with electro-optic devices **100'**, **200**, **300**, **400**, **500**, and/or **600**. Thus, the benefits of such electro-optic devices may be achieved.

[0078] Thus, using method **700**, a controlled chirp may be introduced to a pair of waveguides. The controlled chirp provided in method **700** may compensate for dispersion in optical fibers and/or other components coupled with the waveguides. As a result, the optical signal may be better maintained and longer transmission distances may be achieved. Moreover, inducing controlled

chirp using method 700 may not adversely affect the extinction ratio of the electro-optic device. Consequently, performance of an optical network using method 700 may be improved. [0079] Although the foregoing embodiments have been described in some detail for purposes of clarity of understanding, the invention is not limited to the details provided. There are many alternative ways of implementing the invention. The disclosed embodiments are illustrative and not restrictive.

Claims

1. An electro-optic device, comprising: a first waveguide including a first thin film lithium-containing (TFLC) electro-optic material and being configured to carry a first optical signal; a second waveguide including a second TFLC electro-optic material and being configured to carry a second optical signal; and a plurality of electrodes including a ground electrode, a first signal electrode, and a second signal electrode, the second signal electrode being between the ground electrode and the first signal electrode, the first signal electrode and the second signal electrode being proximate to a portion of the first waveguide and to a portion of the second waveguide; wherein the first signal electrode and the second signal electrode are configured to provide a first modulation to the first optical signal and a second modulation to the second optical signal, a first magnitude of the first modulation being different from a second magnitude of the second modulation such that the electro-optic device has an engineered chirp; and wherein only the portion of the first waveguide is between the first signal electrode and the second signal electrode along an entire length of an electro-optic modulation region.
2. The electro-optic device of claim 1, wherein the plurality of electrodes includes an additional ground electrode, the first signal electrode being between the second signal electrode and the additional ground electrode.
3. The electro-optic device of claim 2, wherein the first signal electrode and the second signal electrode form a differential electrode pair proximate to the portion of the first waveguide and the portion of the second waveguide; and wherein only the portion of the second waveguide is between the second signal electrode and the ground electrode along the entire length of the electro-optic modulation region.
4. The electro-optic device of claim 2, wherein the first signal electrode and the second signal electrode are separated by a first gap proximate to the first waveguide; and wherein the second signal electrode and the ground electrode are separated by a second gap proximate to the second waveguide, the second gap being different from the first gap such that the first modulation has the first magnitude and the second modulation has the second magnitude.
5. The electro-optic device of claim 4, wherein the first gap and the second gap are configured such that the first magnitude (M1) for the first waveguide is twice as large as the second magnitude (M2) for the second waveguide, electro-optic device having chirp parameter (C) of $C=1/3$, with $C=(M1-M2)/(M1+M2)$.
6. The electro-optic device of claim 4, wherein the first gap and the second gap are configured such that the first magnitude (M1) for the first waveguide and the second magnitude (M2) for the second waveguide result in the electro-optic device having a chirp parameter of at least 0.1 and not greater than 0.9 or not greater than -0.1 and at least -0.9, where $C=(M1-M2)/(M1+M2)$.
7. The electro-optic device of claim 4, further comprising: a third waveguide including a third TFLC electro-optic material and being configured to carry a third optical signal; and a fourth waveguide including a fourth TFLC electro-optic material and being configured to carry a fourth optical signal; wherein the plurality of electrodes includes a third signal electrode and a fourth signal electrode proximate to a portion of the third waveguide and to a portion of the fourth waveguide; wherein the first signal electrode and the second signal electrode have a first length proximate to the portion of the first waveguide and the portion of the second waveguide; and

wherein the third signal electrode and the fourth signal electrode have a second length proximate to the portion of the third waveguide and the portion of the fourth waveguide, the first length being different from the second length.

8. The electro-optic device of claim 1, wherein the first waveguide and the second waveguide are coupled to an optical signal transmitter having a dispersion, the engineered chirp being opposite to the dispersion.

9. The electro-optic device of claim 8, wherein the optical signal transmitter includes an optical fiber coupled with the first and second waveguides, the optical fiber having the dispersion.

10. The electro-optic device of claim 1, wherein the first waveguide and the second waveguide are arms of a modulator waveguide including the first waveguide, the second waveguide, a first splitter and a second splitter, the first waveguide and the second waveguide being between the first splitter and the second splitter.

11. An electro-optic device, comprising: a plurality of waveguide pairs, each of the plurality of waveguide pairs including a first waveguide and a second waveguide, each of the plurality of waveguide pairs including a thin film lithium-containing (TFLC) electro-optic material and being configured to carry a first optical signal in the first waveguide and a second optical signal in the second waveguide; and a plurality of electrode trios, each of the plurality of electrode trios including a first signal electrode, a second signal electrode, and a ground electrode, the first signal electrode being proximate to a portion of the first waveguide, the second signal electrode being proximate to a region between the portion of the first waveguide and a portion of the second waveguide, and the ground electrode being proximate to the portion of the second waveguide, the second signal electrode being between the first signal electrode and the ground electrode, the portion of the first waveguide being between the first signal electrode and the second signal electrode; wherein each of the plurality of electrode trios is configured to provide a first modulation having a first magnitude to the first optical signal and a second modulation having a second magnitude to the second optical signal, the first magnitude being different from the second magnitude for at least one of the plurality of electrode trios such that at least one of the plurality of waveguide pairs has an engineered chirp; and wherein only the portion of the first waveguide is between the first signal electrode and the second signal electrode along an entire length of an electro-optic modulation region.

12. The electro-optic device of claim 11, wherein the first signal electrode and the second signal electrode of the at least one of the plurality of electrode trios are separated by a first gap proximate to the portion of the first waveguide of the at least one of the plurality of waveguide pairs; and wherein the second signal electrode and the ground electrode of the at least one of the plurality of electrode trios are separated by a second gap proximate to the second waveguide of the at least one of the plurality of waveguide pairs, the second gap being different from the first gap such that the first modulation has the first magnitude and the second modulation has the second magnitude for the at least one of the plurality of waveguide pairs.

13. The electro-optic device of claim 12, wherein the first gap and the second gap for the at least one of the plurality of electrode trios are configured such that the first magnitude (M1) for the first waveguide of the at least one of the plurality of waveguide pairs and the second magnitude (M2) for the second waveguide of the at least one of the plurality of waveguide pairs result in of the at least one of the plurality of waveguide pairs having a chirp parameter of at least 0.1 and not greater than 0.9 or not greater than -0.1 and at least -0.9, where $C = (M1 - M2) / (M1 + M2)$.

14. The electro-optic device of claim 12, wherein a first electrode trio of the plurality of electrode trios has a first length proximate to a first waveguide pair of the plurality of waveguide pairs and a second electrode trio of the plurality of electrode trios has a second length proximate to a second waveguide pair of the plurality of waveguide pairs, the first length being different from the second length.

15. The electro-optic device of claim 11, wherein each of the plurality of waveguide pairs is

coupled to an optical signal transmitter having a dispersion, the engineered chirp being configured based on the dispersion.

16. The electro-optic device of claim 15, wherein the optical signal transmitter includes an optical fiber coupled to the electro-optic device.

17. A method, comprising: providing a first optical signal to a first waveguide including a first thin film lithium-containing (TFLC) electro-optic material and a second optical signal to a second waveguide including a second TFLC electro-optic material, an electro-optic device including the first waveguide and the second waveguide; and driving electrode signals through a plurality of electrodes including a first signal electrode, a second signal electrode, and a ground electrode, the second signal electrode being between the first signal electrode and the ground electrode, the first signal electrode and the second signal electrode being proximate to a portion of the first waveguide and proximate to a portion of the second waveguide, the electrode signals being configured to provide a first modulation having a first magnitude to the first optical signal and a second modulation having a second magnitude to the second optical signal, the first modulation being different from the second modulation such that the electro-optic device has an engineered chirp; and wherein only the portion of the first waveguide or only the portion of the second waveguide is between the first signal electrode and the second signal electrode along an entire length of an electro-optic modulation region.

18. The method of claim 17, wherein the first signal electrode and the second signal electrode are separated by a first gap proximate to the first waveguide; and wherein the second signal electrode and the ground electrode are separated by a second gap proximate to the second waveguide, the second gap being different from the first gap such that the first magnitude (M1) for the first waveguide and the second magnitude (M2) for the second waveguide result in the electro-optic device having a chirp parameter of at least 0.1 and not greater than 0.9 or not greater than -0.1 and at least -0.9, where $C=(M1-M2)/(M1+M2)$.

19. The method of claim 18, further comprising: driving a third optical signal through a third waveguide including a third TFLC electro-optic material; and driving a fourth optical signal through a fourth waveguide including a fourth TFLC electro-optic material; wherein the plurality of electrodes including a third signal electrode and a fourth signal electrode proximate to a portion of the third waveguide and to a portion of the fourth waveguide; wherein the first signal electrode and the second signal electrode have a first length proximate to the first and second waveguides and the third signal electrode and the fourth signal electrode have a second length proximate to the third and fourth waveguides, the first length being different from the second length.

20. The method of claim 17, wherein the first waveguide and the second waveguide are coupled to an optical signal transmitter having a dispersion, the engineered chirp being opposite to the dispersion.
